flint

Distributed Time Series Analysis Framework For Spark

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Two Sigma

SPARK SUMMIT EUROPE 2016

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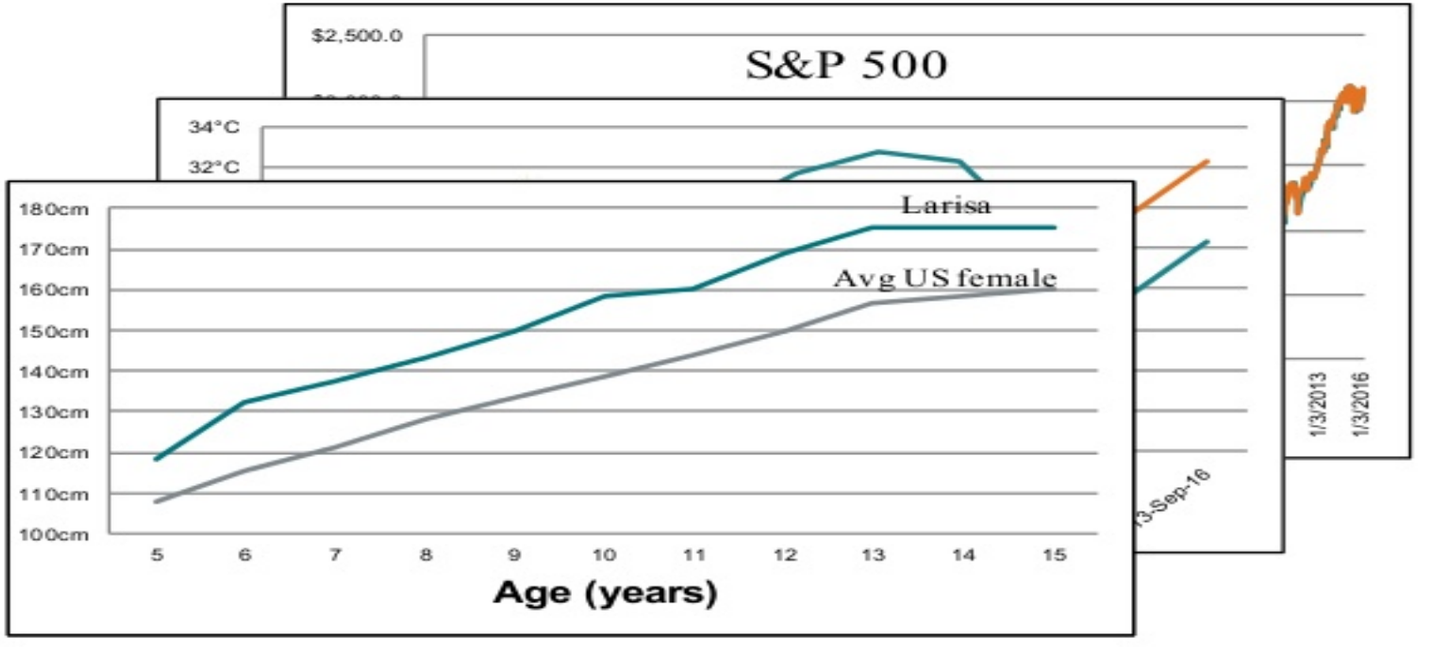
TWO SIGMA

Time series examples

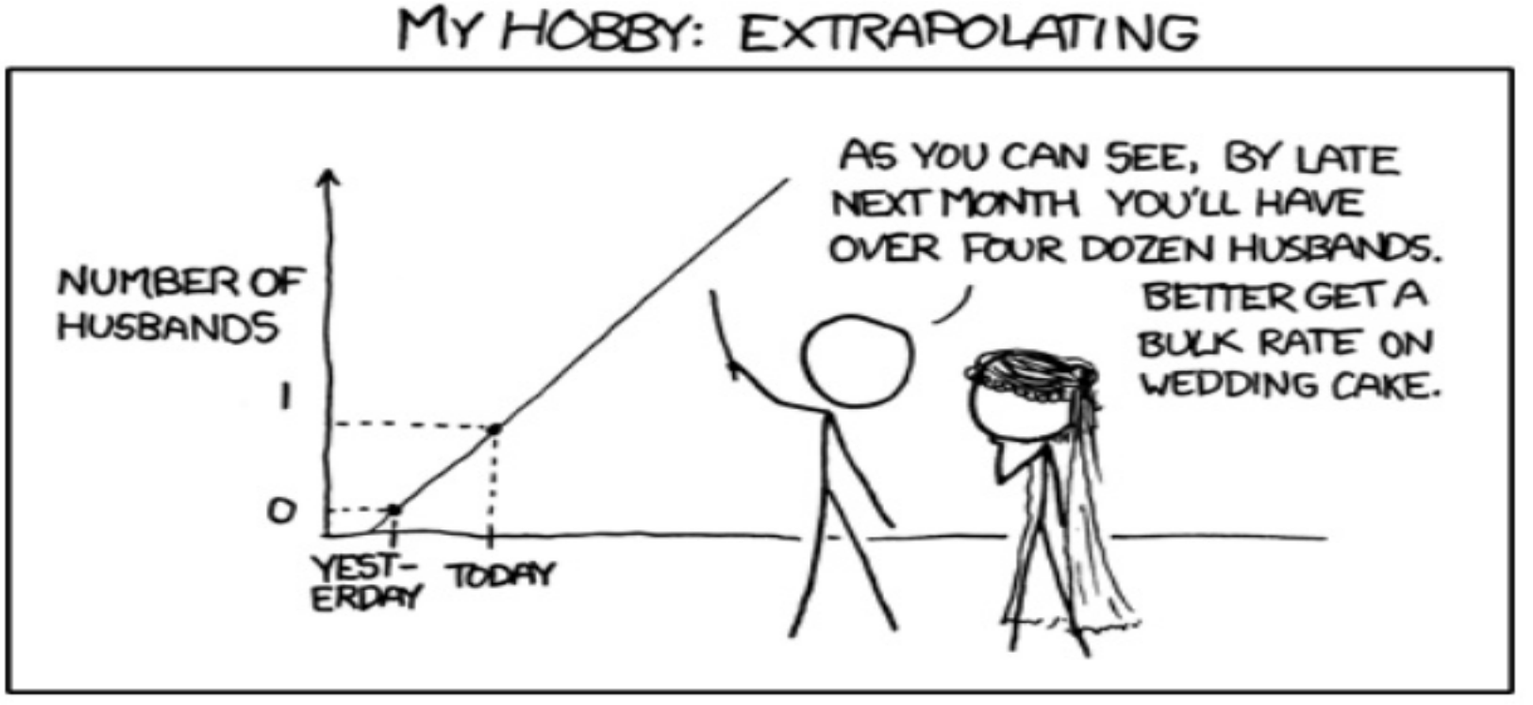
- Stock market prices
- Temperatures
- Height
- …



What do we do with time series data?

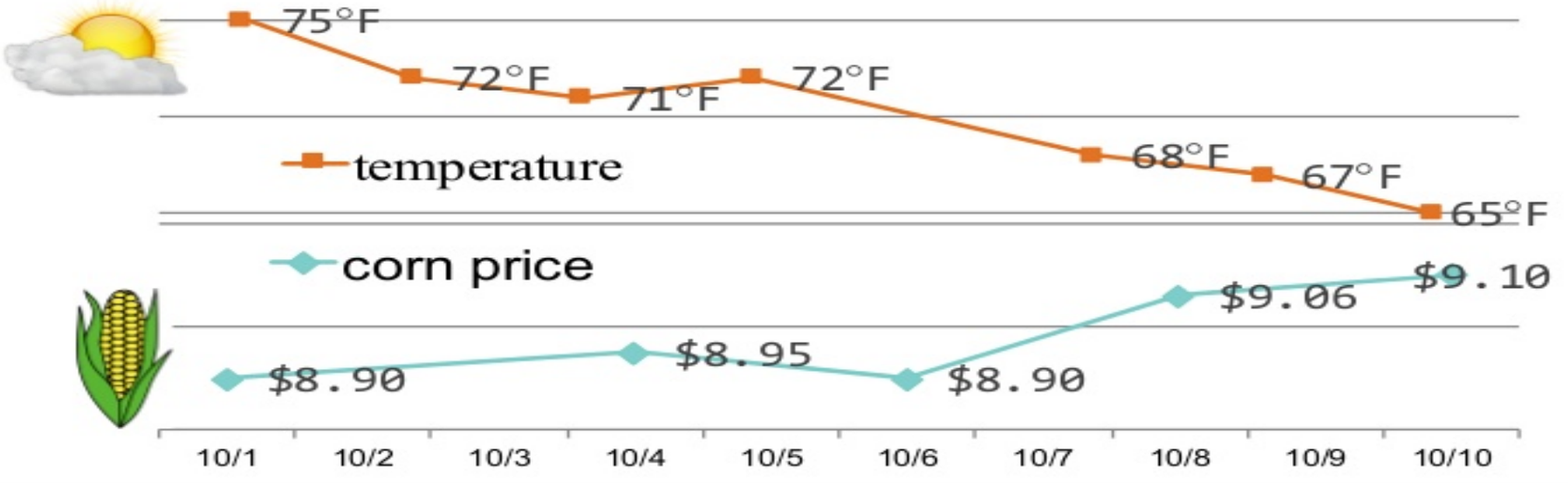
- Forecast future values given past observations

Univariate time series



Multivariate time series

- We can forecast better by joining multiple time series
- Our framework enables fast distributed temporal join of large scale **unaligned** time series
- **Temporal join** is a fundamental operation for time series analysis



Multivariate time series

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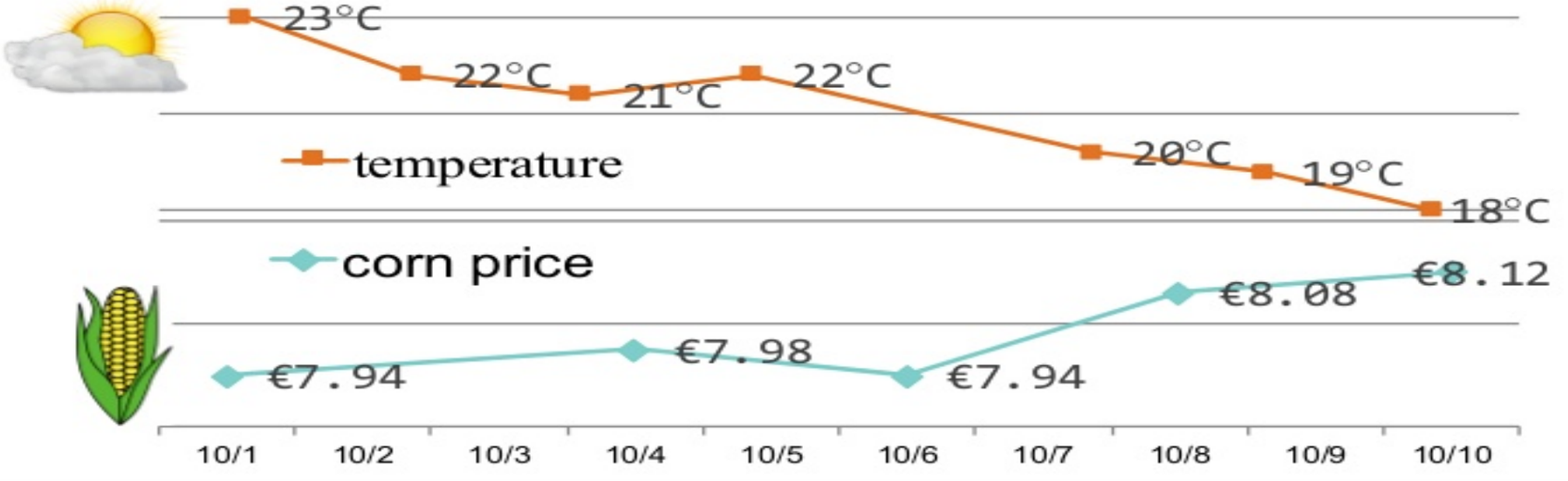
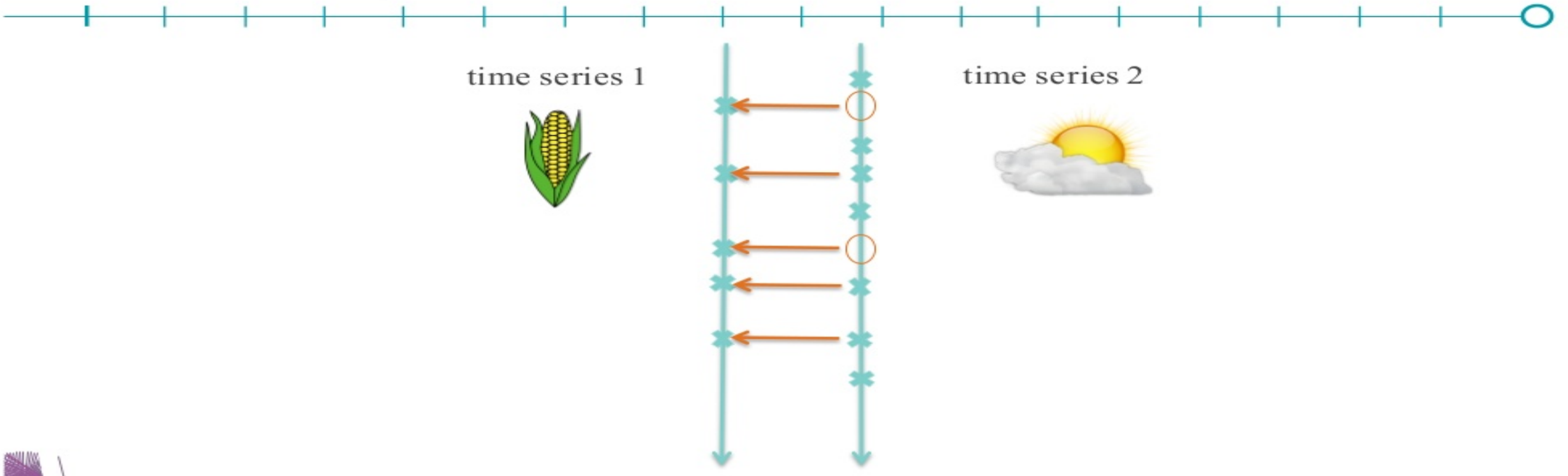
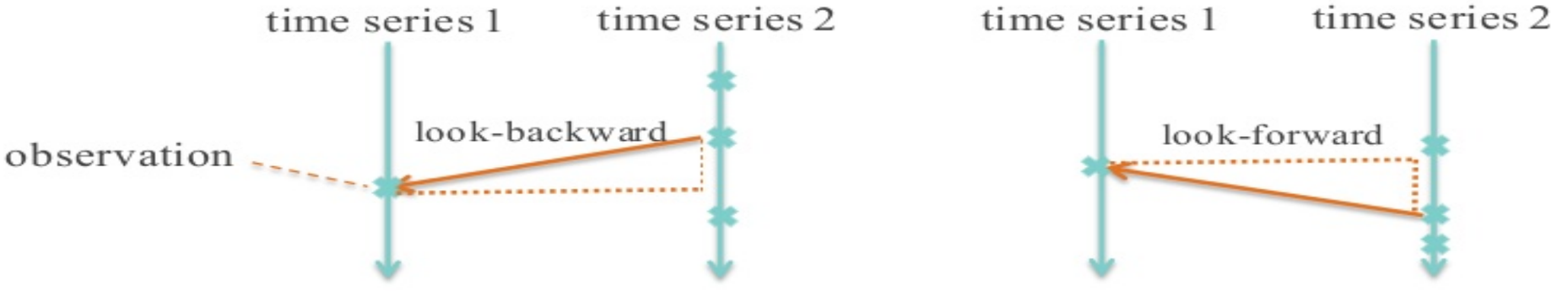


What is a left join?



What is temporal join?

- A particular **join** function defined by a matching criteria over **time**
- Examples of criteria
  - **look-backward**
  - **look-forward**



Temporal join with look-backward criteria

time	tweets
08:00 AM	
10:00 AM	
12:00 PM	

time	BRK.A
08:00 AM	
11:00 AM	

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Temporal joins in practice

time	tweets
08:00 AM	
10:00 AM	
12:00 PM	

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11:00 AM	

We need **fast** and **scalable** distributed temporal join

Existing solutions

- Existing packages don't support temporal join or can't handle large time series
  - Pandas / R / Matlab
    - Limited to single machine
  - Spark
    - Does scale, but all data is unordered
  - spark-ts
    - Expects univariate time series to fit on single machine
    - Splits by col
    - Supports only snapshot data

Flint: A new time series library for Spark

- Goal
  - Provide a collection of functions to **manipulate** and **analyze** time series at scale
    - Group, temporal join, summarize, aggregate …
- How
  - Build a time series aware data structure
    - `TimeSeriesRDD extends RDD`
  - Optimize using temporal locality
    - Reduce shuffling
    - Reduce memory pressure by streaming

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What is a TimeSeriesRDD?

- `TimeSeriesRDD` vs `RDD`
  - Associate time range on each partition
  - Track partition time-ranges
  - Preserve temporal order

RDD

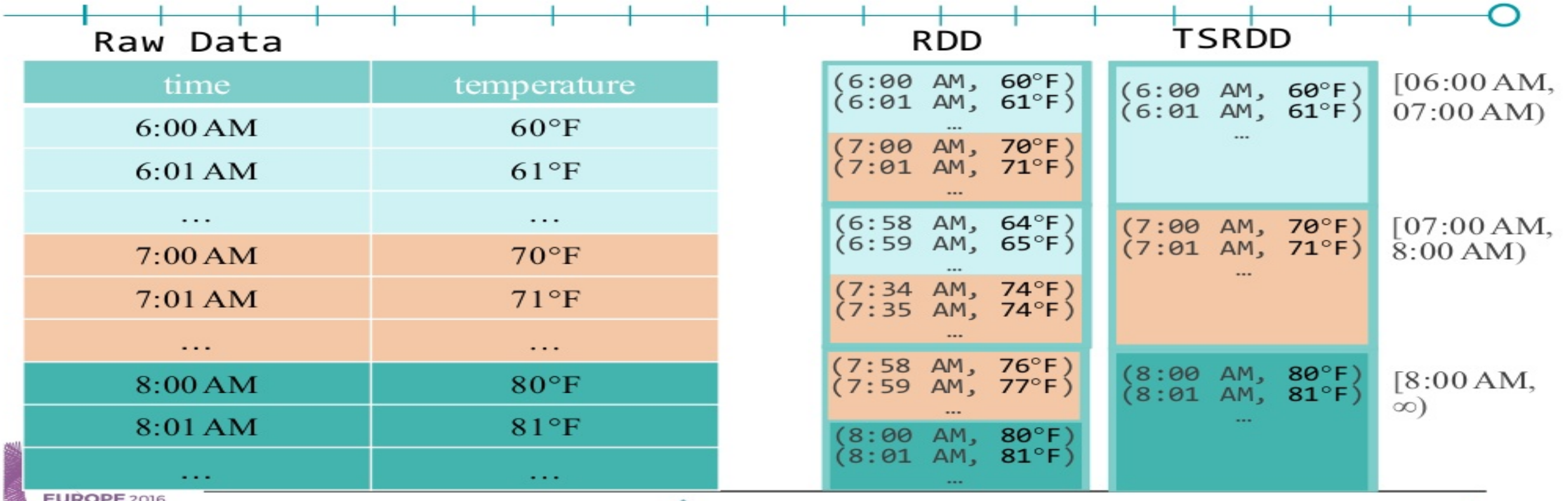
Raw Data

time	temperature
6:00 AM	60°F
6:01 AM	61°F
…	…
7:00 AM	70°F
7:01 AM	71°F
…	…
8:00 AM	80°F
8:01 AM	81°F
…	…

RDD

```
(6:00 AM, 60°F)
(6:01 AM, 61°F)
…
(7:00 AM, 70°F)
(7:01 AM, 71°F)
…
(6:58 AM, 64°F)
(6:59 AM, 65°F)
…
(7:34 AM, 74°F)
(7:35 AM, 74°F)
…
(7:58 AM, 76°F)
(7:59 AM, 77°F)
…
(8:00 AM, 80°F)
(8:01 AM, 81°F)
…
```

TimeSeriesRDD



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Group function

- A **group** function groups rows with exactly the same timestamps

time	city	temperature	
1:00 PM	New York	70°F	group 1
1:00 PM	Brussels	60°F	
2:00 PM	New York	71°F	group 2
2:00 PM	Brussels	61°F	
3:00 PM	New York	72°F	group 3
3:00 PM	Brussels	62°F	
4:00 PM	New York	73°F	group 4
4:00 PM	Brussels	63°F	

Group function

- A **group** function groups rows with nearby timestamps

time	city	temperature
1:00 PM	New York	70°F
1:00 PM	Brussels	60°F
2:00 PM	New York	71°F
2:00 PM	Brussels	61°F
3:00 PM	New York	72°F
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4:00 PM	Brussels	63°F

group 1

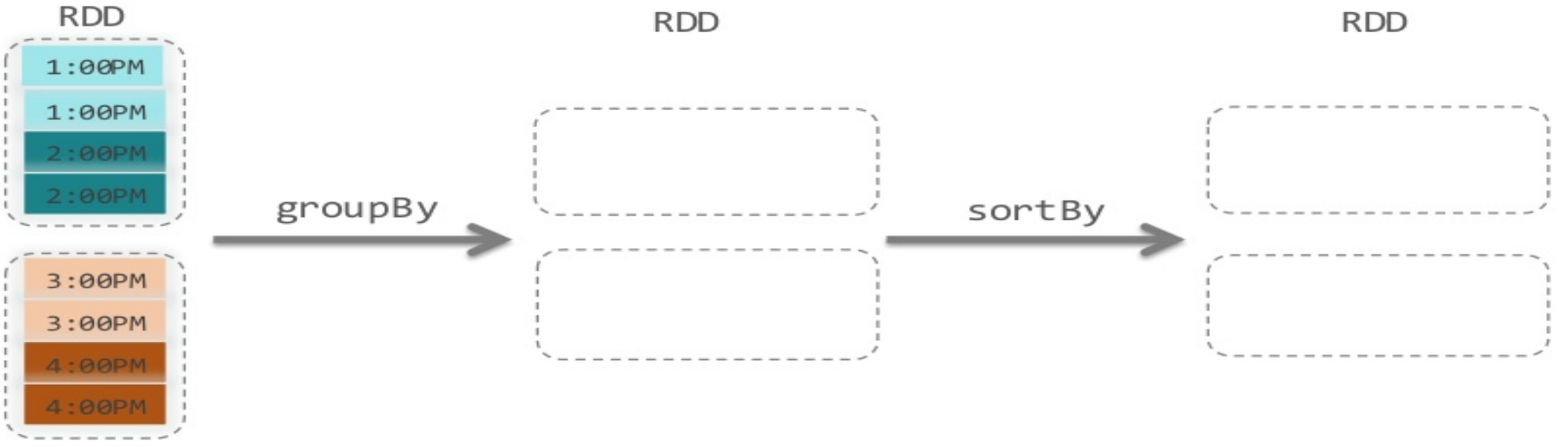
group 2

Group in Spark

- Groups rows with exactly the same timestamps

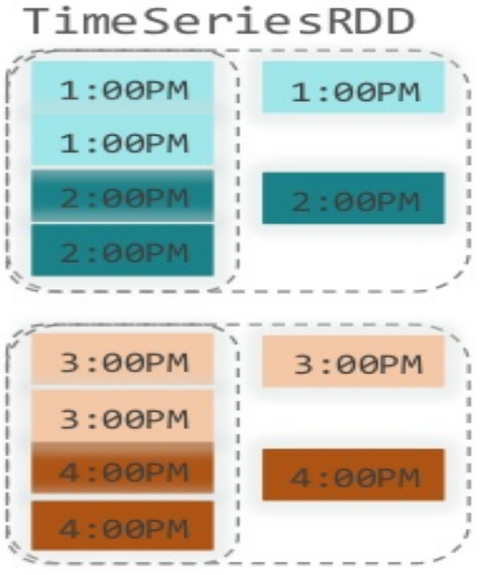
Group in Spark

- Back to Temporal Order
- Data is shuffled and materialized on the workers
  - Temporal order is not preserved



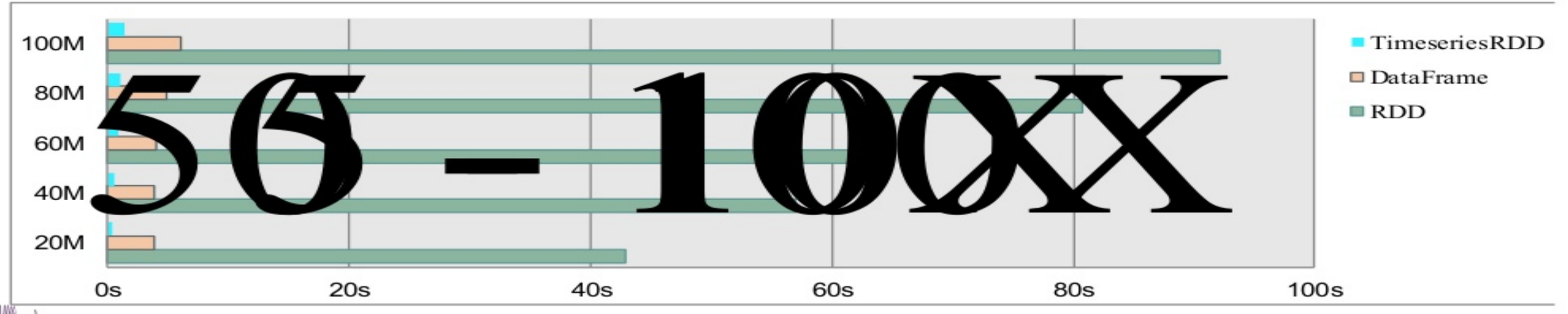
Group in TimeSeriesRDD

- Data is grouped per partition locally as streams



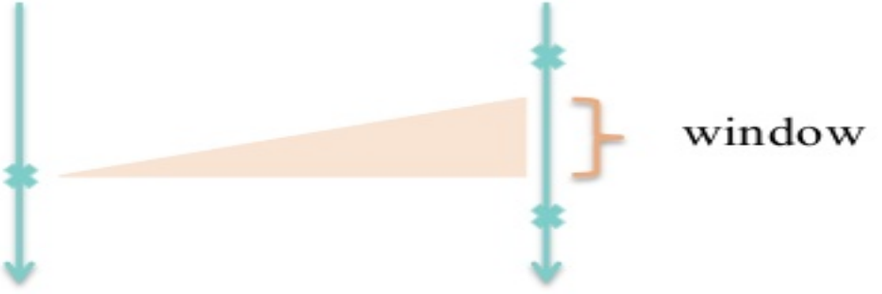
Benchmark for group + count

- Running time of **count** after **group**
  - 16 executors (10G memory and 4 cores per executor)
  - Data read from HDFS



Temporal join

- A **temporal join** function is defined by a matching criteria over **time**
- A typical matching criteria has two parameters
  - **direction** – look-backward or look-forward
  - **window** – how much to look-backward or look-forward



look-backward temporal join

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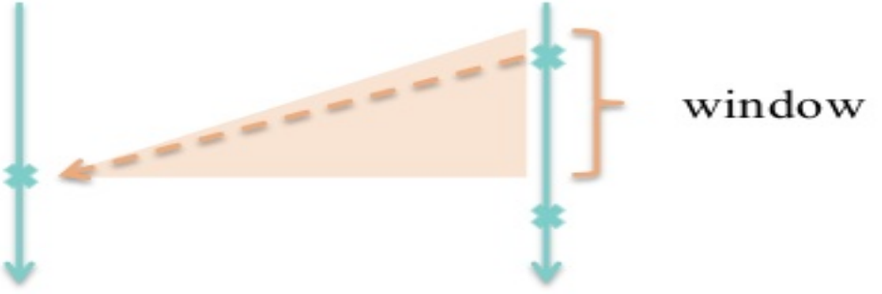
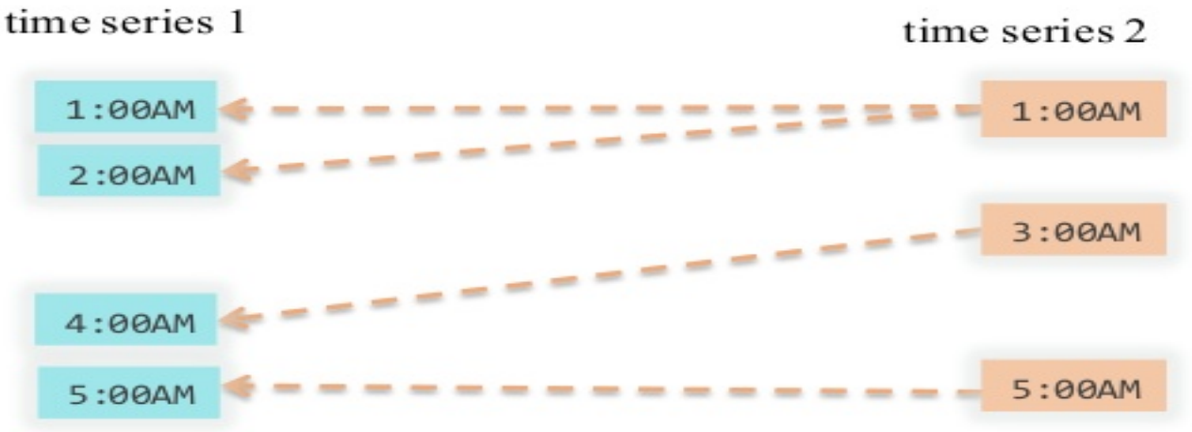


look-backward temporal join

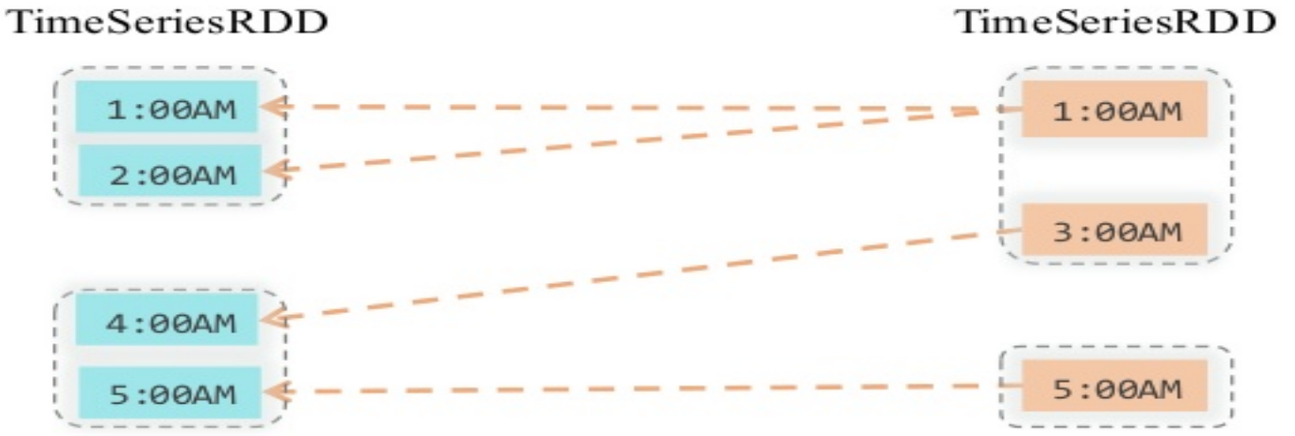
Temporal join

- Temporal join with criteria look-back and window of 1 hour



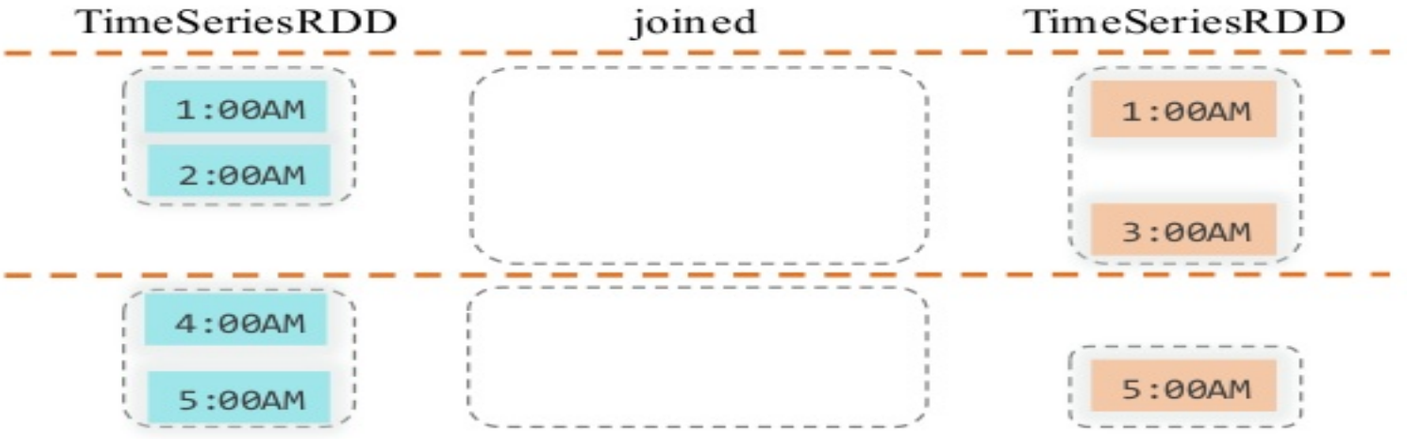
Temporal join

- Temporal join with criteria look-back and window of 1 hour
  - How do we do temporal join in TimeSeriesRDD?



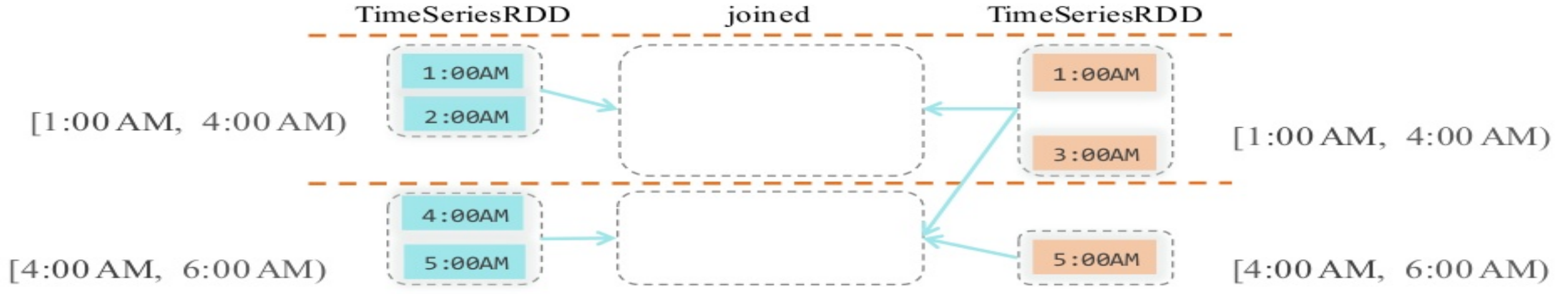
Temporal join in TimeSeriesRDD

- Temporal join with criteria look-back and window of 1 hour
  - partition **time** space into disjoint intervals



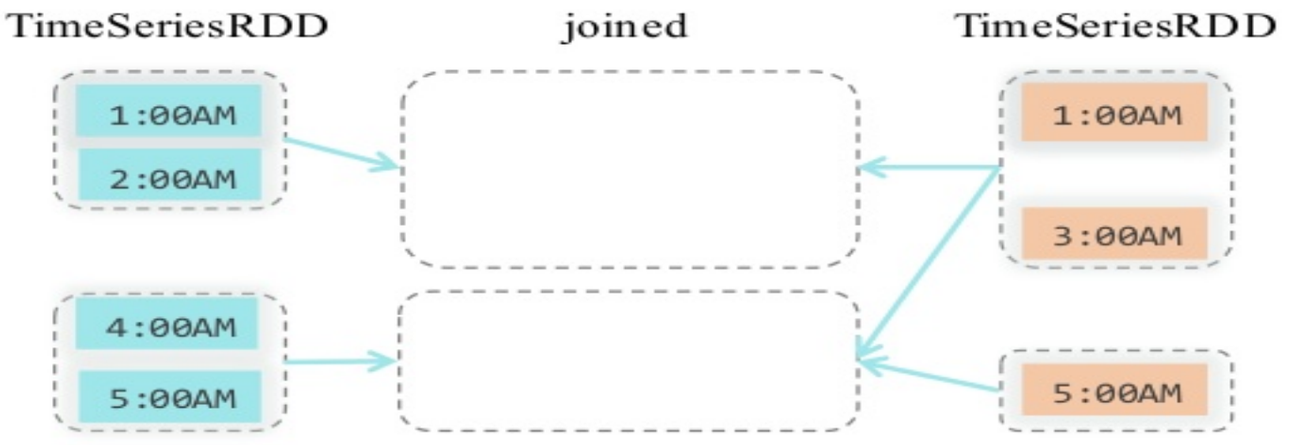
Temporal join in TimeSeriesRDD

- Temporal join with criteria look-back and window of 1 hour
  - Build dependency graph for the joined TimeSeriesRDD



Temporal join in TimeSeriesRDD

- Temporal join with criteria look-back and window of 1 hour
  - Join data as streams per partition



Temporal join in TimeSeriesRDD

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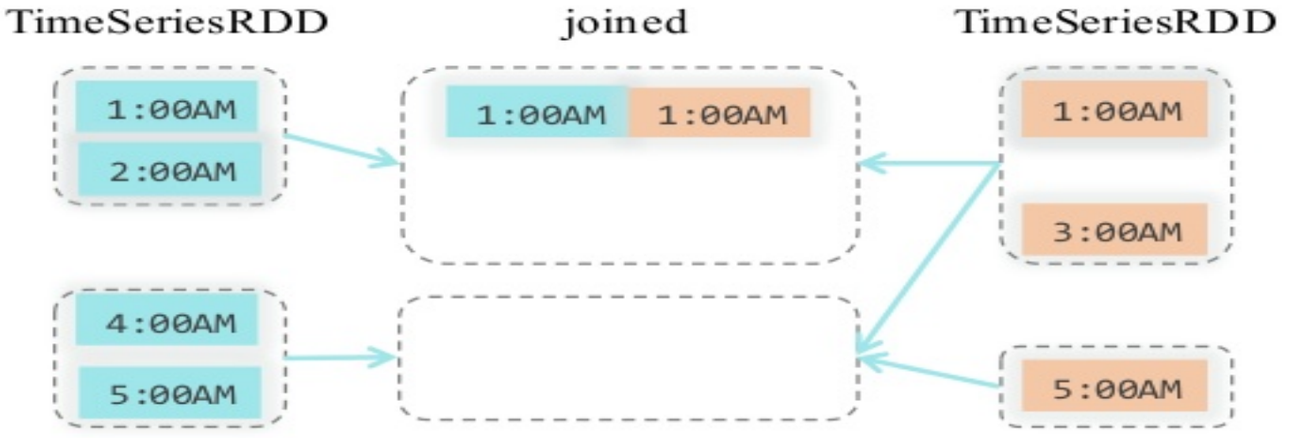

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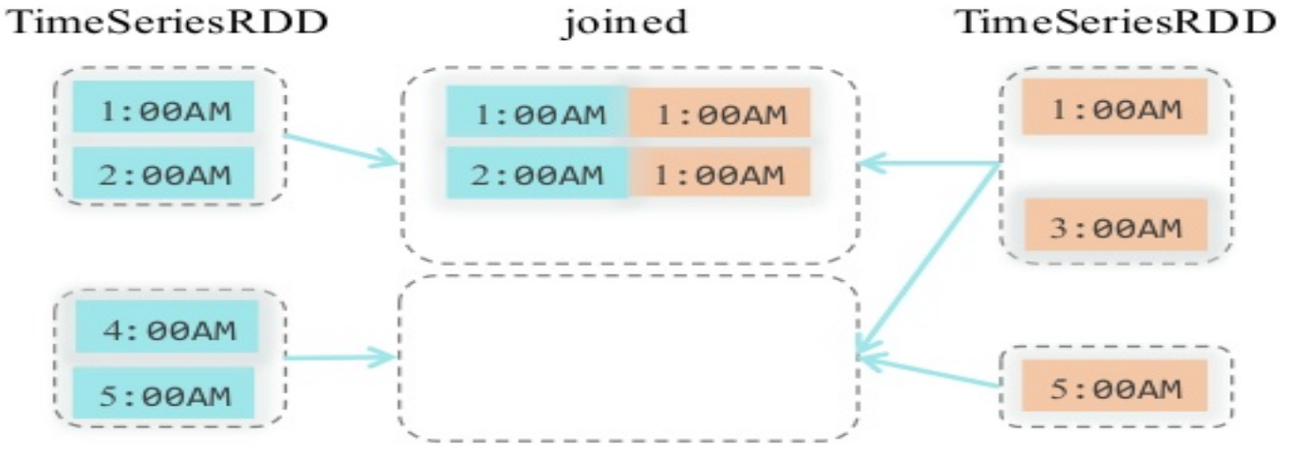

Temporal join in TimeSeriesRDD

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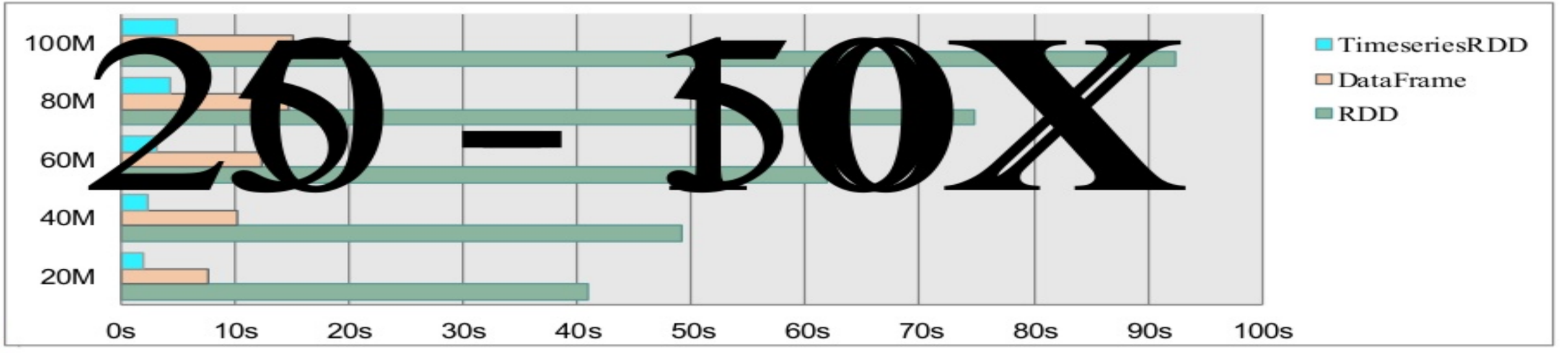

20 – 50X

Functions over TimeSeriesRDD

- Grouping functions
- Temporal joins such as look-forward, look-backward etc.
- Summarizers such as average, variance, z-score etc. over grouping functions

Open Source

- True!
- https://github.com/twosigma/flint

flint

What's next?

- TimeSeriesDataframe / TimeSeriesDataset
- Speed up
  - Richer APIs
- Python bindings
- Additional summarizers

Key contributors

- Christopher Aycock
- Yuri Bogomolov
- Jonathan Coveney
- Li Jin
- David Medina
- Julia Meinwald
- David Palaitis
- Larisa Sawyer
- Leif Walsh
- Wenbo Zhao

flint

Flint: Time Series For Spark

A library to solve for general time series analysis operations at massive scale

Anne Hathaway has nothing to do with Berkshire Hathaway

Check it out in open source, and contribute

https://github.com/twosigma/flint

flint

THANK YOU.

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